

立体几何定理库默写·全覆盖竖式版

一、点线面基本事实与分类

1. 符号约定

$$\left. \begin{array}{l} A \text{ — } l \\ l \text{ — } \alpha \\ \alpha \text{ — } \beta = m \end{array} \right\} \Rightarrow \begin{array}{l} \text{点在线上;} \\ \text{线在面内;} \\ \text{两面交于直线 } m \end{array}$$

2. 位置关系分类

$$\left. \begin{array}{l} \text{线线: 相交、平行、} \text{ — } \\ \text{线面: 在面内、相交、} \text{ — } \\ \text{面面: 平行、} \text{ — } \end{array} \right\} \Rightarrow \text{先判对象, 再选定理}$$

3. 基本事实 1: 三点定面

$$A, B, C \text{ — } \left. \right\} \Rightarrow \text{过 } A, B, C \text{ 有且只有一个平面}$$

4. 基本事实 2: 点线面从属

$$\left. \begin{array}{l} A, B \in l \\ A, B \in \alpha \\ A \neq B \end{array} \right\} \Rightarrow \text{ — }$$

5. 基本事实 3: 公共点推出交线

$$\left. \begin{array}{l} P \in \alpha \\ P \in \beta \\ \alpha \neq \beta \end{array} \right\} \Rightarrow \alpha \cap \beta = \text{ — }, \quad P \in \text{ — }$$

6. 推论 1: 直线与线外一点定面

$$\left. \begin{array}{l} a \text{ 是直线} \\ P \notin a \end{array} \right\} \Rightarrow \text{ — }$$

7. 推论 2: 相交直线定面

$$a \cap b = P \left. \right\} \Rightarrow \text{ — }$$

8. 推论 3：平行直线定面

$$\left. \begin{array}{l} a // b \\ a \neq b \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

二、平行关系定理

9. 平行传递性

$$\left. \begin{array}{l} a // b \\ b // c \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

10. 等角定理

$$\left. \begin{array}{l} a // a' \\ b // b' \end{array} \right\} \Rightarrow \angle(a, b) \text{ 与 } \angle(a', b') \underline{\hspace{2cm}}$$

11. 线面平行判定

$$\left. \begin{array}{l} a // b \\ b \subset \alpha \\ a \not\subset \alpha \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

12. 线面平行性质

$$\left. \begin{array}{l} a // \alpha \\ a \subset \beta \\ \alpha \cap \beta = b \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

13. 面面平行判定

$$\left. \begin{array}{l} a, b \subset \alpha \\ a \cap b = P \\ a // \beta \\ b // \beta \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

14. 面面平行性质

$$\left. \begin{array}{l} \alpha // \beta \\ \gamma \cap \alpha = a \\ \gamma \cap \beta = b \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

15. 面面平行的常用性质

$$\left. \begin{array}{l} \alpha // \beta \\ a \subset \alpha \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

三、垂直关系定理

16. 异面直线垂直定义

$$\left. \begin{array}{l} a' // a \\ a' \cap b = O \\ \angle(a', b) = 90^\circ \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

17. 线面垂直定义推论

$$\left. \begin{array}{l} a \perp \alpha \\ b \subset \alpha \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

18. 线面垂直判定

$$\left. \begin{array}{l} l \perp m \\ l \perp n \\ m, n \subset \alpha \\ m \cap n = P \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

19. 线面垂直性质

$$\left. \begin{array}{l} a \perp \alpha \\ b \perp \alpha \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

20. 面面垂直判定

$$\left. \begin{array}{l} l \perp \alpha \\ l \subset \beta \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

21. 面面垂直性质

$$\left. \begin{array}{l} \alpha \perp \beta \\ \alpha \cap \beta = l \\ a \subset \alpha \\ a \perp l \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

22. 三垂线定理

$$\left. \begin{array}{l} PA \perp \alpha \\ AB \subset \alpha \\ l \subset \alpha \\ AB \perp l \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

23. 三垂线逆定理

$$\left. \begin{array}{l} PA \perp \alpha \\ AB \subset \alpha \\ l \subset \alpha \\ PB \perp l \end{array} \right\} \Rightarrow \underline{\hspace{2cm}}$$

24. 最小角定理

$$\left. \begin{array}{l} l \cap \alpha = A \\ AH \text{ 是 } l \text{ 在 } \alpha \text{ 内的射影} \\ b \subset \alpha, A \in b \end{array} \right\} \Rightarrow \angle(l, AH) \underline{\hspace{1cm}} \angle(l, b)$$

四、距离与角的化归规则

25. 点到直线距离

$$\left. \begin{array}{l} PH \perp l \\ H \in l \end{array} \right\} \Rightarrow d(P, l) = \underline{\hspace{2cm}}$$

26. 点到平面距离

$$\left. \begin{array}{l} PH \perp \alpha \\ H \in \alpha \end{array} \right\} \Rightarrow d(P, \alpha) = \underline{\hspace{2cm}}$$

27. 线面距转点面距

$$\left. \begin{array}{l} a // \alpha \\ P \in a \end{array} \right\} \Rightarrow d(a, \alpha) = \underline{\hspace{2cm}}$$

28. 面面距转点面距

$$\left. \begin{array}{l} \alpha // \beta \\ P \in \alpha \end{array} \right\} \Rightarrow d(\alpha, \beta) = \underline{\hspace{2cm}}$$

29. 异面直线距离

$$\left. \begin{array}{l} M \in a, N \in b \\ MN \perp a \\ MN \perp b \end{array} \right\} \Rightarrow d(a, b) = \underline{\hspace{2cm}}$$

30. 等体积法求点面距

$$\left. \begin{array}{l} V_{P-ABC} = \frac{1}{3} S_{\triangle ABC} \cdot h \end{array} \right\} \Rightarrow h = \underline{\hspace{2cm}}$$

31. 异面直线所成角

$$\left. \begin{array}{l} a' // a \\ b' // b \\ a' \cap b' = O \end{array} \right\} \Rightarrow \angle(a, b) = \underline{\hspace{2cm}}$$

32. 线面角定义

$$\left. \begin{array}{l} l \cap \alpha = A \\ P \in l \\ P \text{ 在 } \alpha \text{ 上的射影为 } H \end{array} \right\} \Rightarrow \angle(l, \alpha) = \underline{\hspace{2cm}}$$

33. 点面距法求线面角

$$\left. \begin{array}{l} \theta = \angle(l, \alpha) \\ P \in l, A = l \cap \alpha \\ d(P, \alpha) = d \\ PA = s \end{array} \right\} \Rightarrow \sin \theta = \underline{\hspace{2cm}}$$

34. 二面角平面角定义

$$\left. \begin{array}{l} \alpha \cap \beta = l \\ O \in l \\ OA \subset \alpha, OB \subset \beta \\ OA \perp l, OB \perp l \end{array} \right\} \Rightarrow \text{_____ 是二面角 } \alpha - l - \beta \text{ 的平面角}$$

35. 二面角面积射影法

$$S_{\text{射影}} = S_{\text{原}} \cos \theta \quad \left. \vphantom{S_{\text{射影}}} \right\} \Rightarrow \cos \theta = \text{_____}$$